

Section C: General Information

1 Introduction

- 1.1 Rationale
- 1.2 Certification title
- 1.3 Language
- 1.4 Exclusions
- 1.5 Key Skills
- 1.6 Code of Practice Requirements
- 1.7 Spiritual, Moral, Ethical, Social and Cultural Issues
- 1.8 Environmental Education, European Dimension and Health and Safety Issues
- 1.9 Avoidance of Bias
- 1.10 Calculators and computers

2 Specification Aims

- 2.1 Aims of MEI
- 2.2 Aims of this specification

3 Assessment Objectives

- 3.1 Application to AS and A2
- 3.2 Specification Grid

4 Scheme of Assessment

- 4.1 Units of Assessment
- 4.2 structure
- 4.3 Rules of Combination
- 4.4 Final Certification
- 4.5 Availability
- 4.6 Re-sits
- 4.7 Question Papers
- 4.8 Coursework
- 4.9 Special Arrangements
- 4.10 Differentiation
- 4.11 Grade Descriptions

5 Subject Content

- 5.1 Assumed Knowledge
- 5.2 Modelling
- 5.3 Competence Statements

6 Unit Specifications

- 6.1 Introduction to Advanced Mathematics, C1 (4751) AS
- 6.2 Concepts for Advanced Mathematics, C2 (4752) AS
- 6.3 Methods for Advanced Mathematics, C3 (4753) A2
- 6.4 Applications of Advanced Mathematics, C4 (4754) A2
- 6.5 Further Concepts for Advanced Mathematics, FP1 (4755) AS
- 6.6 Further Methods for Advanced Mathematics, FP2 (4756) A2
- 6.7 Further Applications of Advanced Mathematics, FP3 (4757) A2
- 6.8 Differential Equations, DE (4758) A2
- 6.9 Mechanics 1, M1 (4761) AS
- 6.10 Mechanics 2, M2 (4762) A2
- 6.11 Mechanics 3, M3 (4763) A2
- 6.12 Mechanics 4, M4 (4764) A2
- 6.13 Statistics 1, S1 (4766) AS
- 6.14 Statistics 2, S2 (4767) A2
- 6.15 Statistics 3, S3 (4768) A2
- 6.16 Statistics 4, S4 (4769) A2
- 6.17 Decision Mathematics 1, D1 (4771) AS
- 6.18 Decision Mathematics 2, D2 (4772) A2
- 6.19 Decision Mathematics Computation, DC (4773) A2
- 6.20 Numerical Methods, NM (4776) AS
- 6.21 Numerical Computation, NC (4777) A2
- 6.22 Further Pure Mathematics with Technology, FPT (4798) A2

7 Further Information and Training for Teachers

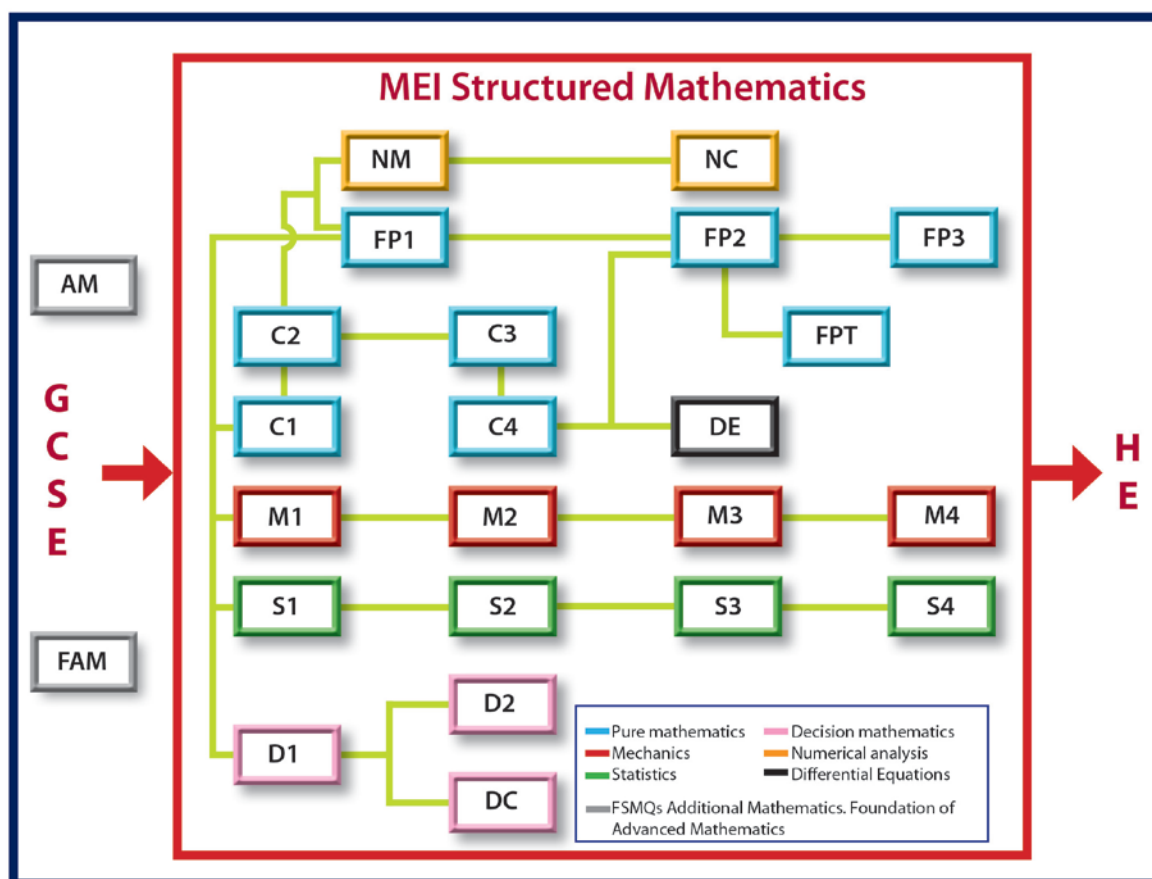
Appendix A: Mathematical Formulae

Appendix B: Mathematical Notation

Vertical black lines indicate a significant change to the previous printed version.

SECTION A: SPECIFICATION SUMMARY

The units in MEI Structured Mathematics form a step-by-step route of progression through the subject, from GCSE into the first year of university. For those who are insecure about their foundation, access to the scheme is provided by the Free Standing Mathematics Qualification, *Foundations of Advanced Mathematics*.



The subject is developed consistently and logically through the 22 AS and A2 units, following strands of Pure Mathematics, Mechanics, Statistics, Decision Mathematics and Numerical Analysis. Each unit is designed both as a worthwhile and coherent course of study in its own right, taking about 45 hours of teaching time, and as a stepping stone to further work.

Suitable combinations of three and six modules give rise to AS and Advanced GCE qualifications in Mathematics, Further Mathematics and Pure Mathematics. Candidates usually take their units at different stages through their course, accumulating credit as they do so.

The normal method of assessment is by unit examinations, held in June each year, in most cases lasting 1½ hours. Three units also have coursework requirements. Candidates are allowed to re-sit units, with the best mark counting.

The Advanced Subsidiary GCE is assessed at a standard appropriate for candidates who have completed the first year of study of a two-year Advanced GCE course, i.e. between GCSE and Advanced GCE. It forms the first half of the Advanced GCE course in terms of teaching time and content. When combined with the second half of the Advanced GCE course, known as 'A2', the advanced Subsidiary forms 50% of the assessment of the total Advanced GCE. However the Advanced Subsidiary can be taken as a stand-alone qualification. A2 is weighted at 50% of the total assessment of the Advanced GCE.

SECTION B: USER SUMMARY

USING THIS SPECIFICATION

This specification provides a route of progression through mathematics between GCSE and Higher Education and/or employment.

- Students start with AS Mathematics. This consists of the two AS units in Pure Mathematics, *C1* and *C2*, together with one applied unit, either *M1*, *S1* or *D1*.
- Many students take one year over AS Mathematics but this is not a requirement; they can take a longer or a shorter time, as appropriate to their circumstances.
- Examinations are available in June only.
- Unit results are notified in the form of a grade and a Uniform Mark. The total of a candidate's Uniform Marks on relevant modules determines the grade awarded at AS GCE (or Advanced GCE).
- A unit may be re-sat any number of times with the best result standing.
- To obtain an AS award a 'certification entry' must be made to OCR. There is no requirement for candidates going on to Advanced GCE to make such an entry.
- To complete Advanced GCE Mathematics, candidates take three more units, *C3*, *C4* and another applied unit.
- The applied unit may be in the same strand as that taken for AS in which case it will be an A2 unit (e.g. *S2* following on from *S1*). Alternatively it may be in a different strand in which case it will be an AS unit (e.g. *M1* following on from *S1*).
- Many candidates will take these three units in the second year of their course but there is no requirement for this to be the case.
- An Advanced GCE award will only be made to those who apply for it.
- Candidates may also take Further Mathematics at AS and Advanced GCE. There is considerable flexibility in the way that this can be done.
- AS Further Mathematics consists of *FP1* and two other units which may be AS or A2.
- The three units for AS Further Mathematics may be taken in the first year. The compulsory unit *FP1* has been designed to be accessible for students who have completed Higher Tier GCSE and are studying *C1* and *C2* concurrently. The AS units, *M1*, *S1*, *D1* and *NM* are also suitable for those taking AS Further Mathematics in the first year.
- Many of those who take AS Further Mathematics in the first year then take another three units in their second year to obtain Advanced GCE Further Mathematics. Such candidates take 12 units, six for Mathematics and six for Further Mathematics.
- Other AS Further Mathematics students spread their study over two years rather than completing it in the first year.
- Another pattern of entry is for candidates to complete Advanced GCE Mathematics in their first year and then to go onto Further Mathematics in their second year.
- Those who take Advanced GCE Mathematics and AS Further Mathematics must do at least 9 units.

- The rules of aggregation mean that it is to candidates' advantage to certificate Advanced GCE Mathematics and AS or Advanced GCE Further Mathematics at the same time. This can be achieved, if necessary, by recertificating for any previously entered qualification in any series when an entry for a qualification is made.
- Candidates who take 15 or 18 units are eligible for additional awards in Further Mathematics.

SUMMARY OF CHANGES FROM THE PREVIOUS MEI SPECIFICATION

The revisions to the subject criteria have resulted in considerable changes to the assessment arrangements compared to those in the previous specification (those for first teaching in September 2000). These in turn have affected the provision of units in MEI Structured Mathematics and their content:

- the core material is now covered in four units, two at AS and two at A2. The two AS units are compulsory for Advanced Subsidiary Mathematics and all four core units are compulsory for Advanced GCE Mathematics;
- consequently the first four units in the Pure Mathematics strand, *C1* to *C4* are all new;
- only two applied units now contribute to Advanced GCE Mathematics;
- the reduction in the amount of Applied Mathematics in Advanced GCE Mathematics means that it is no longer feasible to provide as many applied units for Further Mathematics, and so there are fewer Mechanics and Statistics units. However the provision in Decision Mathematics remains unaltered;
- *Mechanics 4* and *Statistics 4* are new units, drawing material from a number of units in the previous specification;
- in addition there are some changes to *Statistics 1*, *2* and *3*; these reflect their new status within the Advanced GCE, particularly the fact that *Statistics 3* is no longer the natural ending point for those Advanced GCE students whose Applied Mathematics is entirely statistics;
- Advanced Subsidiary Further Mathematics may now be obtained on **three** AS units. One of these is a new Pure Mathematics unit *Further Concepts for Advanced Mathematics, FP1*;
- the content of *Further Concepts for Advanced Mathematics, FP1*, depends only on the AS subject criteria and so is inevitably different from that of the unit which it replaces. This change has had knock-on effects to the content of the two remaining units in Pure Mathematics;
- the number of subject titles available has been reduced in line with the new subject criteria.

In addition, there are some other changes that are not a direct consequence of the new subject criteria. In particular, those responsible for this specification were aware that, following the introduction of Curriculum 2000, mathematics was making much greater demands on students' time than other subjects and so there is a reduction in the amount of coursework required.

SECTION C: GENERAL INFORMATION

1 Introduction

1.1 RATIONALE

1.1.1 This Specification

This booklet contains the specification for MEI Structured Mathematics for teaching from September 2013. It covers Advanced Subsidiary GCE (AS) and Advanced GCE (A Level) qualifications in Mathematics and Further Mathematics and also in Pure Mathematics.

This specification was developed by Mathematics in Education and Industry (MEI) and is assessed by OCR. Support for those delivering the specification comes from both bodies and this is one of its particular strengths.

This specification is designed to help more students to fulfil their potential by taking and enjoying mathematics courses that are relevant to their needs post-16. This involves four key elements: breadth, depth, being up-to-date and providing students with the ability to use their mathematics.

- Most students at this level are taking mathematics as a support subject. Their needs are almost as diverse as their main fields of study, and consequently this specification includes the breadth of several distinct strands of mathematics: Pure Mathematics, Mechanics, Statistics, Decision Mathematics and Numerical Analysis.
- There are, however, those students who will go on to read mathematics at university and perhaps then become professional mathematicians. These students need the challenge of taking the subject to some depth and this is provided by the considerable wealth of Further Mathematics units in this specification, culminating in *FP3*, *M4* and *S4*.
- Mathematics has been transformed at this level by the impact of modern technology: the calculator, the spreadsheet and dedicated software. There are many places where this specification either requires or strongly encourages the use of such technology. The units *DC*, *NC* and *FPT* have computer based examinations; an option in *FP3* is based on graphical calculators, and the coursework in *C3* and *NM* is based on the use of suitable devices.
- Many students complete mathematics courses quite well able to do routine examination questions but unable to relate what they have learnt to the world around them. This specification is designed to provide students with the necessary interpretive and modelling skills to be able to use their mathematics. Modelling and interpretation are stressed in the papers and some of the coursework and there is a comprehension paper as part of the assessment of *C4*.

MEI is a curriculum development body and in devising this specification the long term needs of students have been its paramount concern.

This specification meets the requirements of the Common Criteria (QCA, 1999), the GCE Advanced Subsidiary and Advanced Level Qualification-Specific Criteria (QCA, 1999) and the Subject Criteria for Mathematics (QCA, 2002).